ICOHP Descriptors and Thermodynamic Stability Across a 349-Structure Dataset

Campaign 2: Dataset Construction, VASP+LOBSTER Computation, and Statistical Analysis

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The project's central question is viability discrimination based on Δ(ICOHP) (§6.4), across a 349-structure dataset.

Contents

1 Objective and Problem Statement

Central research question (see §6.4): does Δ(ICOHP) — the reaction-ICOHP descriptor for a compound's decomposition into elements, and specifically the sign-based endobondic/exobondic classification it supports — distinguish thermodynamically stable, metastable, and unstable compounds?

2 Methodology

2.1 Dataset Selection

Materials Project query: unary and binary chemical systems, non-magnetic (|total magnetization| $\leq$ 0.01 μ_B/formula unit — MP's categorical `magnetic_ordering` field is "Unknown" for ∼98% of entries and would have excluded nearly every genuinely non-magnetic compound), no f-block elements. The dataset combines two complementary selection tracks: a first set of 186 compounds ($\leq$ 20 atoms/cell, 60 per stability family, a 2-entries-per-chemical-system cap to ensure elemental diversity rather than many polymorphs of the same system), and a second set of 163 compounds/elements ($\leq$ 40 atoms/cell) specifically targeting additional elemental references, off-equilibrium polymorphs (high-pressure phases, allotropes), and alkali/alkaline-earth binaries against N/O/F/P/S/Cl, chosen to build genuine same-composition polymorph groups and to test descriptors far from equilibrium.

2.2 Bonding-Type Classification

Bond type (ionic / covalent / metallic) is computed by default by the `classify()` function (elemental composition + `is_metal`). A second, independent classification is built directly from each compound's ICOHP/ICOBI data (§5): `is_metal` for metallic character, then the mean ICOBI per bond (bond order, covalency) to separate ionic from covalent among non-metallic compounds. The two classifications are compared in §5.

2.3 VASP + LOBSTER Computation

Same template across all structures (static run, `ISYM=`−1, `LASPH=.TRUE.`, PAW-PBE potentials recommended by `pymatgen.io.vasp.sets.MPRelaxSet`): `NBANDS` is derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`), and basis-function determination goes through `Lobsterin.get_basis(potcar_symbols=...)` rather than direct POTCAR-file parsing (whose internal pymatgen validity check fails on several POTCARs unrecognized by hash in the local mirror).

2.4 ICOHP/ICOBI Quantities: Integration, Cutoffs, Units

LOBSTER generation (`lobsterin`, identical across the dataset): `pbeVaspFit2015` basis; COHP integration window from −35 to +5 eV around the Fermi level (`COHPstartEnergy`/`COHPendEnergy`); bond detection by interatomic distance from 0.1 to 6.0 Å (`cohpGenerator`, orbital-resolved); 0.05 eV Gaussian smearing (`gaussianSmearingWidth`).

Integrated ICOHP/ICOBI (`ICOHPLIST.lobster`/ `ICOBILIST.lobster`). A bond's integrated ICOHP is, by LOBSTER's own definition, the integral of COHP(E) from the bottom of the window above up to the Fermi level (occupied states only), in eV; negative = bonding, positive = antibonding. ICOBI is the same integral over COBI (bond order index), dimensionless. These are the per-bond, then per-compound-aggregated, quantities feeding §5 and the classic correlations of §6.

Reaction Δ(ICOHP) (`reaction_analysis/`). For a balanced reaction (typically compound → elements), Δ = sum of products' ICOHP − sum of reactants' ICOHP, "products −

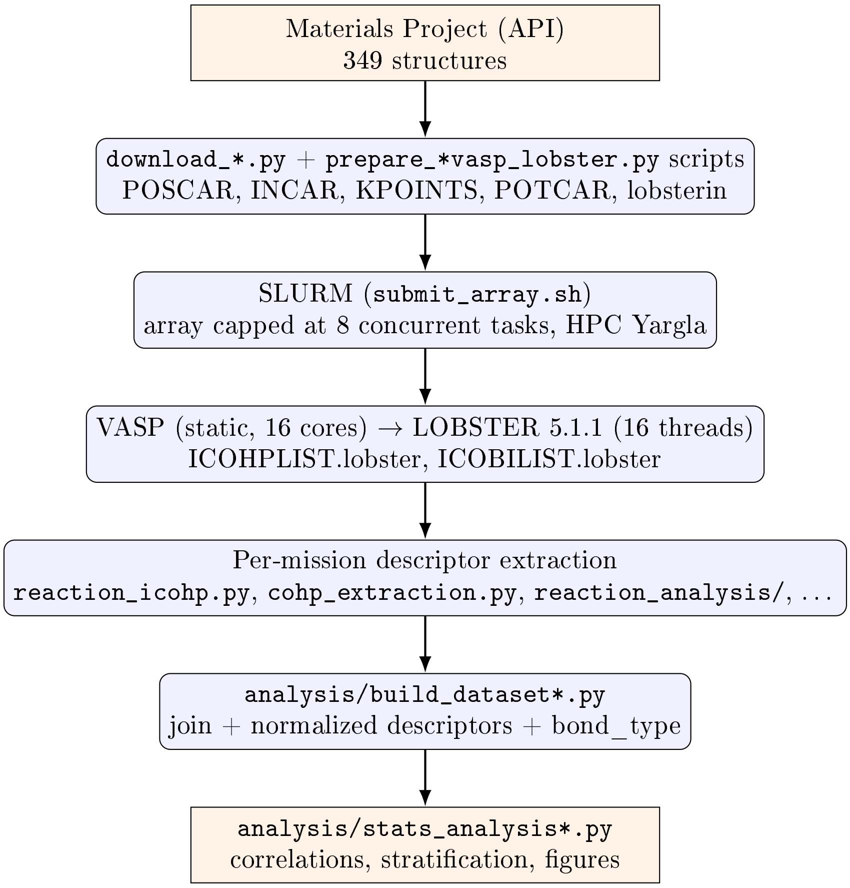
reactants" convention, in three normalizations: per formula unit (extensive), per transferred atom (the normalization used throughout this report unless noted), and per bond (diagnostic only, non-conservative — bonds are not conserved between differently-bonded structures). Positive sign ("endobondic"): breaking the reactant's bonds costs more than the products' bonds recover — a possible kinetic barrier. Negative sign ("exobondic"): no such barrier is visible from bonding alone.

Antibonding population near the Fermi level (`cohp_extraction.py`) — a quantity distinct from the Δ(ICOHP) above. Rather than integrating the whole occupied COHP, this quantity integrates only its antibonding part (COHP>0) over a one-sided window $(E_{\mathrm{ref}} - \Delta E,\ E_{\mathrm{ref}}]$, where $E_{\mathrm{ref}} = E_F$ (metallic compounds) or the VBM (gapped compounds), and $\Delta E \in \{0.5, 1.0, 2.0\}$ eV ($\Delta E = 1.0$ eV as the primary window) — only occupied states can destabilize the ground state via antibonding, hence the one-sided rather than symmetric window. Reported both raw (same "eV" unit convention as LOBSTER, though it is a Hamilton population integral, not literally an energy) and normalized by the total occupied ICOHP magnitude at the same E_{ref}. This quantity, and only this one, feeds the descriptor in §6.2 — the reaction Δ(ICOHP) (§6.3–6.4) integrates the whole occupied COHP with no restriction to a window near E_F.

2.5 Statistics

Spearman correlation (no reason to expect a linear relationship) between each ICOHP/ICOBI descriptor and a stability target (`energy_above_hull` or `formation_energy_per_atom` depending on the descriptor), overall and per subgroup (`bond_type`, `is_metal`); any $n < 15$ subgroup is explicitly flagged, never presented bare. **No SISSO, no symbolic regression** at this stage (see §9).

3 Workflow Diagram



4 Environment

Identical across all 349 structures (Yargla, VASP 6.5.0, LOBSTER 5.1.1, Python 3.11 `.venv`), extended with: `mp-api`, `pandas`, `scipy`, `scikit-learn`, `matplotlib`, `pydantic`, `pyyaml` (see `requirements.txt`). Per-step core allocation: §8.

5 Dataset

The full set of computed compounds and elements totals **349 structures**.

Table 1: Dataset composition by element count.

Composition	n
Single-element references	68
Multi-element compounds	281
Total	**349**

Bond type. Two independent classifications (§2.2): the `classify()` function (elemental composition + `is_metal`) and a classification built directly from each compound's ICOHP/ICOBI data (`is_metal` for metallic character, then a threshold on mean ICOBI per bond — calibrated at the midpoint of the median ICOBI of compounds `classify()` already labels ionic/covalent, i.e. 0.0381 — to separate ionic from covalent among non-metallic compounds).

Table 2: Stability class (by E_{hull} and experimental/theoretical character).

Class	n	E_{hull} range (eV/atom)
stable ($E_{\text{hull}} = 0$)	124	0.0000
experimental metastable	119	0.000035 – 2.0568
theoretical metastable	106	0.000084 – 3.1688
Total	**349**	

Table 3: Bond type: `classify()` vs. the ICOHP/ICOBI classification (concordance on the subset `classify()` labels: 157/174 = 90.2%).

`classify()` \ ICOHP/ICOBI	metallic	ionic	covalent	unclassified
metallic	88	0	0	0
ionic	0	48	5	0
covalent	0	12	21	0
unclassified	102	36	34	1
Total	**190**	**96**	**60**	**1**

Of the 173 compounds `classify()` leaves "unclassified" (essentially alkali/alkaline-earth × {N, P, S} pairs, outside `IONIC_ANIONS`), the ICOHP/ICOBI classification resolves **70** into ionic (36) or covalent (34); the remaining 103 are either metallic (102, already covered by `is_metal`) or lack ICOBI data (1). Where both methods render a verdict, agreement is strong (90.2%) but not total: 10 ionic/covalent disagreements, worth checking case by case before using either classification as a stratification variable.

6 Results

6.1 Correlations (Spearman) — classic ICOHP aggregates

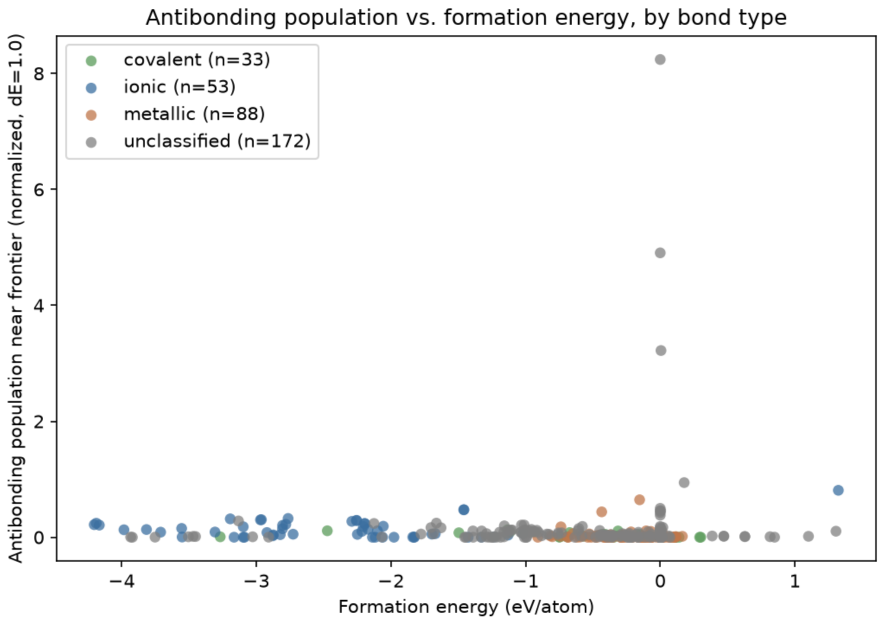
Table 4: Spearman correlations, classic ICOHP aggregates (sum, minimum) against E_{hull}, 186-compound subset.

Group	Metric	n	ρ	p
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

These four rows serve as a reference baseline: raw ICOHP aggregates (per-compound sum, minimum) show little to no correlation with E_{hull}, except within the metallic subgroup. The descriptors built further on (antibonding population, reaction ICOHP, §6.2–§6.4) do markedly better, against `formation_energy_per_atom` rather than `energy_above_hull`.

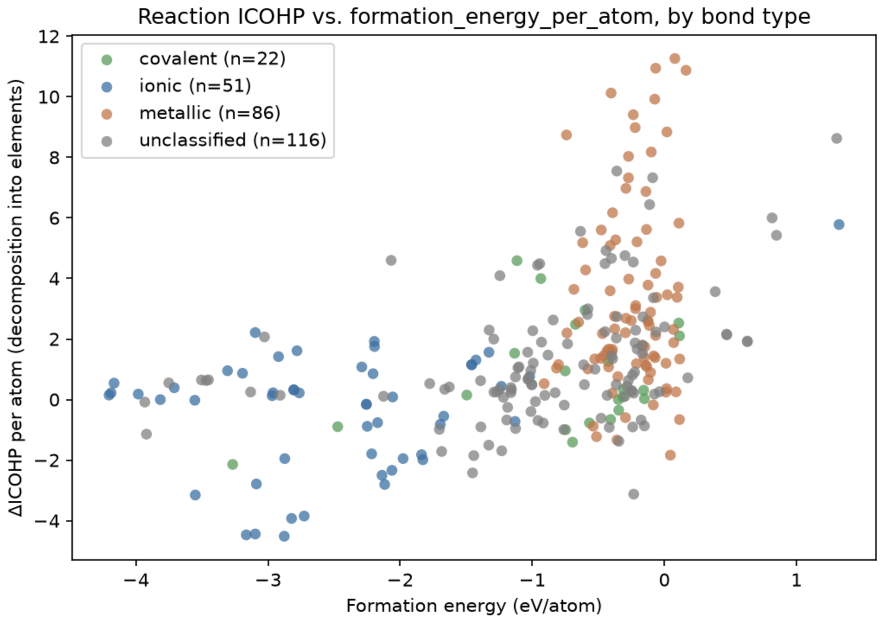
6.2 Antibonding Population Near the Fermi Level

A question distinct from the fully-integrated ICOHP (§2.4): not *how much* bonding a compound has, but *how* its COHP is distributed in energy — the occupied antibonding fraction just below the Fermi level/VBM, by analogy with Peierls/Jahn–Teller-type electronic instabilities. Result, against `formation_energy_per_atom`: $\rho = -0.282$, $p = 9.2 \times 10^{-8}$ ($n = 346$). The sign is the *opposite* of the naive Peierls/Jahn–Teller reading. The `bond_type=covalent` subgroup ($n = 33$) survives stratification clearly ($\rho = -0.551$, $p = 0.0009$) — the project's most robust stratified result; `is_metal=False` also survives ($n = 157$, $p = 0.001$). An earlier result on the `bond_type=ionic` subgroup ($n = 9$, $\rho = -0.683$) does not replicate once the sample grows to $n = 53$ ($\rho = -0.184$, not significant) — a clear demonstration of the danger of $n < 15$ subgroups.



6.3 Reaction ICOHP

An ICOHP analog of `formation_energy_per_atom`: is a compound's total ICOHP "cheaper" than that of the same atoms as reference elements? Result: $\rho = 0.502$, $p = 6.3 \times 10^{-19}$ ($n = 275$) — the dataset's strongest and most robust continuous correlation, with an intuitive sign (less bonding than in elemental form $\Rightarrow$ less stable formation). `bond_type` stratification fully explains this layer of the pooled signal (Kruskal–Wallis $p = 1.1 \times 10^{-15}$, no stratum survives) — **but `is_metal` stratification survives on both sides** (`is_metal=False`: $\rho = 0.319$, $p = 0.0001$; `is_metal=True`: $\rho = 0.256$, $p = 0.0025$): between-group clustering fully explains the `bond_type`-level grouping, but not a real relationship within a fixed `is_metal`. Case 2 (direct same-formula polymorph comparison, 49 groups): 53.1% agreement between the most-bonded and the most-stable member, against a 45.9% chance baseline (exact Poisson-binomial test, $P = 0.19$) — no signal.



Still no SISSO. Reaction ICOHP is the dataset's strongest raw correlate of `formation_energy_per_atom`, and the only descriptor whose signal is not fully explained away by between-group clustering at every level tested — the best argument yet for a stratified (`is_metal`-conditioned) rather than pooled feature search, still not attempted. Detailed reports: `analysis/REPORT_antibonding.md`, `analysis/REPORT_reaction_icohp.md`.

6.4 Δ(ICOHP)-Based Viability Discrimination

Full detail: `analysis/REPORT_delta_icohp_viability.md`, `analysis/test_delta_icohp_viability.py`.

Reference validation. Per-bond-type ICOHP values for 7 reactions with independently known ΔICOHP (transcribed independently of any calculation in this project) are run end to end through `balance.py`/`delta.py`/`classify.py`:

Table 5: Validating the endobondic/exobondic classification on 7 reactions with independently known ΔICOHP.

Reaction	Reference (kJ/mol)	Computed (kJ/mol)	Label Ref.	Label Computed
$Pb(N_3)_2 \rightarrow Pb + 3\,N_2$	+1345	+1344.3	endo	endo
$S_4N_2 \rightarrow \frac{1}{2}S_8 + N_2$	+258	+258.5	endo	endo
$S_4N_4 \rightarrow \frac{1}{2}S_8 + 2N_2$	+399	+397.3	endo	endo
ZnSn → Zn + Sn	−337	−336.6	exo	exo
CaO[sphalerite] → CaO[rocksalt]	−79	−78.9	exo	exo
$CaN \rightarrow \frac{1}{3}Ca_3N_2 + \frac{1}{6}N_2$	−205	−204.8	exo	exo
$Mn_2O_7 \rightarrow 2MnO_2 + \frac{3}{2}O_2$	−186	−187.7	exo	exo

The 7 cases reproduce the reference ΔICOHP to within a few kJ/mol (source per-bond value rounding) and every `BondingLabel` matches. The Mn_2O_7 case is a useful guardrail on its own: exobondic under this static sign criterion, yet Mn_2O_7 is a real, observed compound that simply decomposes slowly (gradual O_2 loss) rather than through abrupt bond breaking — `classify.py` structurally attaches a warning to *every* `UNSTABLE_NONEXISTENT` verdict for exactly this reason, rather than special-casing Mn_2O_7.

Correlation alone (Pearson/Spearman) is not the right test for "do our numbers match the reference values" — it is satisfied by any monotonic relationship, even with a large systematic offset or scale factor. **Lin's concordance correlation coefficient** (CCC) specifically tests agreement with the identity line (computed = reference), the claim actually being tested here:

Table 6: Computed-vs-reference concordance on the 7 validation reactions.

Statistic	Value
n	7
Lin's CCC	**0.999998**
Pearson r	0.999999 ($p = 3.5 \times 10^{-15}$)
Sign agreement	7/7
Mean |residual|	0.76 kJ/mol
Max |residual|	1.70 kJ/mol
RMSE	0.98 kJ/mol

A CCC this close to 1 means the two methods agree essentially to the reporting precision of the reference values themselves (rounded to 2–3 decimals), not merely that they move together — the implementation is validated, not just spot-checked.

Applied to the full dataset, pooled. 281 case-1 reactions, every element and compound with a computed decomposition-into-elements reaction, pooled into a single population (grouping by selection provenance would be exactly the kind of between-group confound this project's methodology is meant to catch).

Table 7: Spearman correlations, ΔICOHP/atom (case 1, 281 reactions) against two continuous stability targets.

Target	n	ρ	p
`formation_energy_per_atom`	275	−0.5016	6.3×10^{-19}
`energy_above_hull_eV_per_atom`	279	−0.1096	0.068

(Sign: `reaction_analysis`'s products−reactants convention, opposite to `reaction_icohp.py`'s own column §6.3 — same magnitude, flipped sign.) The formation-energy result is the same relationship as §6.3's, confirmed here on the pooled population.

Sign-based discrimination (endobondic/exobondic). Experimental viability indicator (`theoretical=False` means experimentally realized):

Table 8: Endobondic/exobondic label against experimental/theoretical status.

	endobondic	exobondic
experimental ($n = 177$)	47	130
theoretical only ($n = 98$)	17	81

Fisher's exact test: $p = 0.10$, odds ratio = 1.72 (endobondic is $\sim 1.7\times$ more frequent among experimentally realized compounds — the expected direction, endobondic being designed as a kinetic-barrier signature — but not significant at this sample size).

Kruskal–Wallis across the three groups: $H = 2.81$, $p = 0.245$ — **not significant**, but the ordering is exactly what physics predicts: experimentally metastable compounds (known to exist despite sitting off the convex hull — exactly the category the endobondic/exobondic framework addresses) have the highest endobondic fraction; never-synthesized theoretical metastable compounds have the lowest.

Table 9: Median ΔICOHP/atom and % endobondic by stability class ($n = 175$).

Class	n	median ΔICOHP/atom (eV)	% endobondic
experimental metastable	59	−1.077	**27.1%**
stable	59	−1.589	20.3%
theoretical metastable	59	−1.659	**11.9%**

Reading. The 7 reference reactions reproduce almost exactly — the implementation is correct. Extending the same sign-based logic to a much larger, more heterogeneous, and generally much more comfortably stable population does not yet produce a statistically significant stable/metastable/unstable discriminant, even though every trend points the right way. This is not a contradiction: the 7 reference reactions were specifically chosen as stability edge cases ($Pb(N_3)_2$, S_4N_2/S_4N_4, ZnSn, Mn_2O_7) where a kinetic-barrier explanation is specifically needed; most of this project's 281 compounds are not — they are comfortably stable or comfortably not, where the thermodynamic driving force (the formation-energy correlation above) can simply dominate the sign-based kinetic signal. The full `ViabilityLabel` machinery in `classify.py` (beyond the sign-based `BondingLabel` alone used here) needs exactly such a near-stability-boundary population — for the current dataset, decomposition into elements is strongly exothermic for almost every compound (normal for stable inorganic chemistry), which would trivially classify nearly everything `STABLE_ON_HULL` and add no information beyond what `energy_above_hull` already says directly.

7 Failures Found and Fixed

Per the instruction ("every job failure must be logged with its probable cause"), 5 of the first 178 SLURM-array-submitted jobs failed on the first pass, all diagnosed and fixed at the source (the fixes, applied to the shared SLURM/INCAR templates, carry over to the whole dataset):

- **$Al_{12}W$, BW, WO_2, TcW, TiW** (all tungsten-containing): the local mirror's `W_sv` POTCAR is missing its `LEXCH` field entirely (confirmed by direct inspection — present in every other variant checked, including plain `W`), which VASP reads as an incompatible exchange-correlation functional and refuses to run. Fixed by replacing the `W_sv` override with plain `W` (valid PBE, verified).
- **$Al_{12}W$** failed again after that fix, this time with a VASP SIGSEGV — a known issue (`ulimit -s unlimited` before `vasp_std`), present in `submit_array.sh` but forgotten in `prepare_vasp_lobster.py`'s per-compound SLURM template. Fixed; recovered on retry.

Final success rate across all 349 structures computed to date: **349/349**.

8 Compute Cost

349 structures computed in total: 68 single-element references and 281 multi-element compounds (§5).

Every VASP job, then LOBSTER job, runs on the **Yargla** HPC cluster, within a single SLURM allocation of one node (`-nodes=1`) and **16 cores**: VASP under MPI parallelism (`-ntasks=16 -cpus-per-task=1`, `srun`), then LOBSTER under OpenMP parallelism on the same 16 cores (`OMP_NUM_THREADS=16`, single process, run sequentially after VASP within the same job). The SLURM array (`submit_array.sh`) is capped at 8 concurrent tasks.

Combined sequential total (VASP + LOBSTER, 349 structures): **55.3 h**. Given the 8-concurrent-task cap, the order of magnitude for actual elapsed time to process this volume as a

Table 10: VASP and LOBSTER compute time across the dataset.

	VASP	LOBSTER
n	349	348*
Mean (s)	168	404
Median (s)	58	110
Max (s)	5715 (1.6 h)	12063 (3.4 h)
Sequential total	16.3 h	39.0 h

* 1 of 349 structures had no usable LOBSTER step (trivial-cell reference element).

single batch would be ∼6.9 h — a rough estimate (dividing by 8, ignoring queue time) given for context.

9 Current Limits and Next Steps

- **`classify()` coverage**: 173/349 structures (50%) fall outside the ionic/covalent/metallic categories, mostly alkali/alkaline-earth × {N, P, S} bonds that `IONIC_ANIONS` does not recognize (§5) — purely informative here, but a real limitation for any mission that would want to stratify on it.
- **No true dynamical "unstable" class**: `theo_metastable` is a proxy (never synthesized), not a phonon-calculation result.
- **Per-subgroup sample size**: remains this project's structural limit at every new stratification (`bond_type`, `is_metal`) — see §6.2 for a concrete example where an $n < 15$ subgroup did not replicate once the sample grew.
- **Why not SISSO yet**: reaction ICOHP (§6.3) is now the strongest correlate and the first whose signal is not fully explained away by between-group clustering at every level tested — the best argument yet for a stratified feature search, still not attempted.

10 Conclusion

Reitz & Dronskowski's endobondic/exobondic methodology (§6.4) reproduces faithfully across the 7 independently known reference reactions (Lin's CCC 0.999998) and applies to this report's full dataset (281 decomposition-into-elements reactions). On that population, sign-based ΔICOHP discrimination points the expected way (endobondic more frequent among experimental compounds, and among metastable rather than stable or never-synthesized compounds) but does not reach statistical significance at this scale (§6.4). Reaction ICOHP otherwise remains the dataset's strongest and most robust continuous correlate against formation energy ($\rho = 0.502$, §6.3).

Two concrete directions follow. First, deliberately extending to compounds near the stability boundary, where a kinetic-barrier explanation is specifically needed, rather than letting the dataset grow incidentally from whatever other analyses require — most current compounds are comfortably stable or comfortably unstable, a regime where the thermodynamic driving force simply dominates the sign-based kinetic signal. Second, a new descriptor: a reaction Δ(ICOHP) restricted to a window near the Fermi level, modeled on the antibonding population of §6.2 (§2.4) rather than on the fully integrated ICOHP — since it is precisely the near-E_F window that carries the project's most robust stratified subgroup, a "near-E_F" version of the reaction ΔICOHP is a natural, not-yet-attempted extension.

Source code: https://github.com/John-Akapulco/viability
Data: `analysis/percolation_vs_formation_energy.csv`
Detailed reports: `analysis/REPORT_*.md`

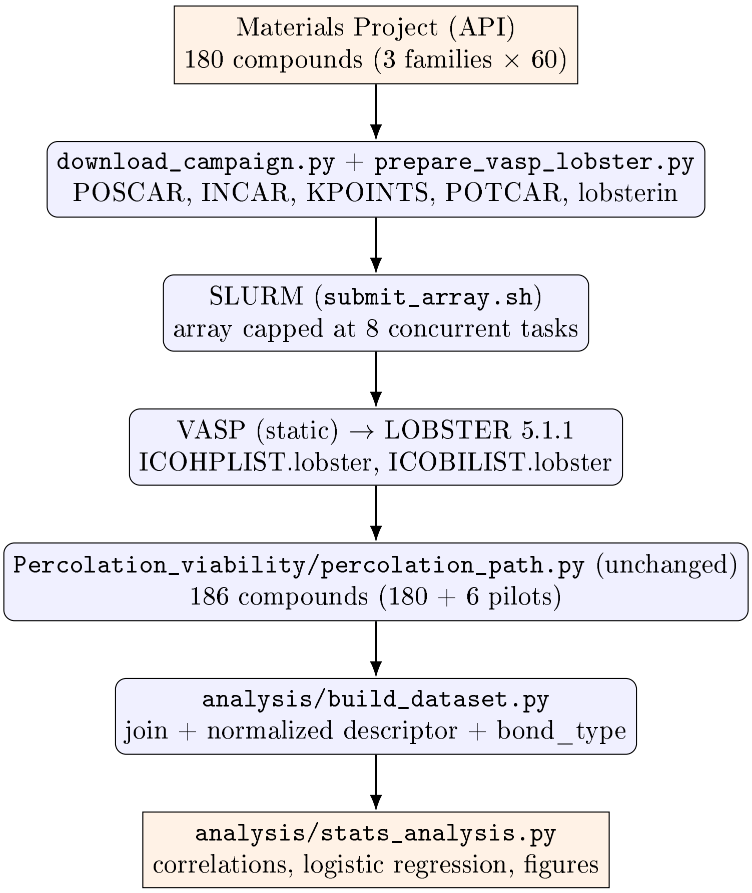
A Appendix: Percolation_viability

This appendix collects everything related to the percolation weight and its direct descendants (network dimensionality, periodic min-cut — missions #1 and #3): this was the central subject of the original campaign 2, unmodified below from its first drafting, but it is no longer this project's organizing question (§6.4). The corresponding source code has been moved to the repository's `Percolation_viability/` directory (`percolation_path.py`, `network_dimensionality.py`, `periodic_mincut.` and their tests `tests/test_percolation_path.py`, `tests/test_network_dimensionality.py`, `tests/test_periodic_mincut.py`).

A.1 Original objective (mission #1)

The very first report (6 toy compounds) established that the percolation descriptor computed by `percolation_path.py` diverges sharply from classic ICOHP aggregates (sum, mean, min, max), but without a sample large enough to statistically test whether that difference is *useful*: does the percolation descriptor carry predictive information about thermodynamic stability (`energy_above_hull`) that the classic aggregates don't already carry? Campaign 2 built a 186-compound dataset (180 new + the 6 pilot compounds) split into three families (`exp_stable`, `exp_metastable`, `theo_metastable`, §5) to settle that question.

A.2 Original workflow diagram



A.3 Campaign 2 dataset (186 compounds)

Family	n	E_{hull} range (eV/atom)
exp_stable	63	0.0000 (by construction)
exp_metastable	63	0.0005 – 0.0907
theo_metastable	60	0.0013 – 0.1991

Bond type (`classify()`)	n
metallic	88
covalent	23
ionic	9
unclassified	66

35% of compounds (66/186) fall outside `classify()`'s three categories — mostly transition-metal pnictides/chalcogenides and hydrogen-containing compounds. (Historical snapshot of campaign 2 alone; the up-to-date table across the current 349 structures is in §5.)

A.4 Results (with the percolation weight)

Group	Metric	n	ρ	p
all	percolation weight (raw)	186	0.064	0.383
all	percolation weight (normalized)	186	0.071	0.339
metallic	percolation weight (raw)	88	0.191	0.075
metallic	percolation weight (normalized)	88	**0.222**	**0.038**
ionic*	percolation weight (raw)	9	−0.402	0.284
covalent	percolation weight (raw)	23	0.165	0.453
all	ICOHP sum	186	0.032	0.670
all	ICOHP min	186	0.093	0.209
metallic	ICOHP sum	88	0.223	0.037
metallic	ICOHP min	88	0.198	0.065

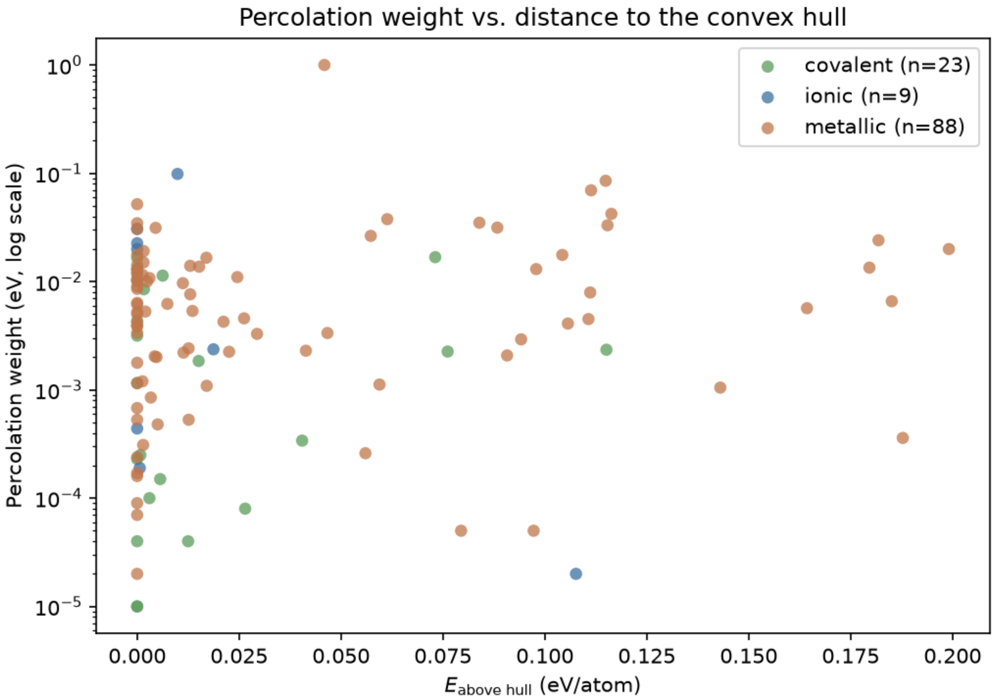
* $n < 15$: do not over-interpret.

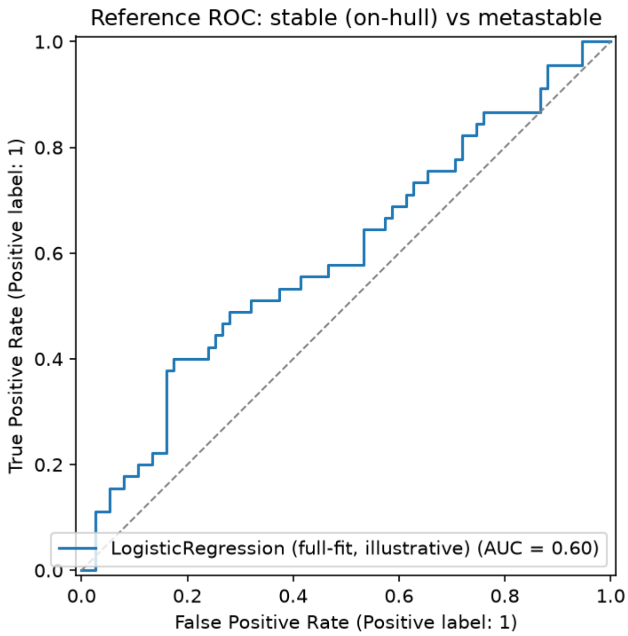
In the one subgroup where any result reaches nominal significance (metallic, $n = 88$), the normalized percolation weight ($\rho = 0.222$, $p = 0.038$) and the ICOHP sum ($\rho = 0.223$, $p = 0.037$) are statistically indistinguishable in strength. **The percolation descriptor does not demonstrate additional predictive power over the classic aggregates on this dataset**.

Reference logistic regression. Stable (on-hull) vs. metastable, features: normalized percolation weight, mean ICOHP, bond-type dummies ($n = 120$ after dropping unclassified compounds).

AUC (5-fold cross-validation)	0.496 ± 0.141
per-fold AUC	0.556, 0.430, 0.252, 0.637, 0.607

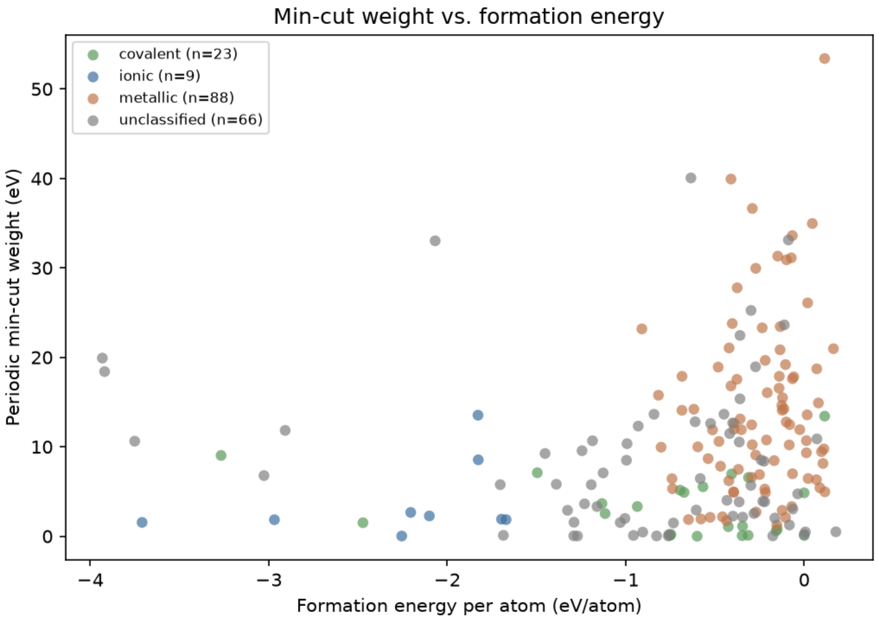
AUC ≈ 0.5: chance level. The wide spread across folds (0.25–0.64) is itself informative: it indicates the absence of a stable, learnable signal at this sample size, not merely a "weak but real" one.





A.5 Mission #3: network dimensionality and periodic min-cut

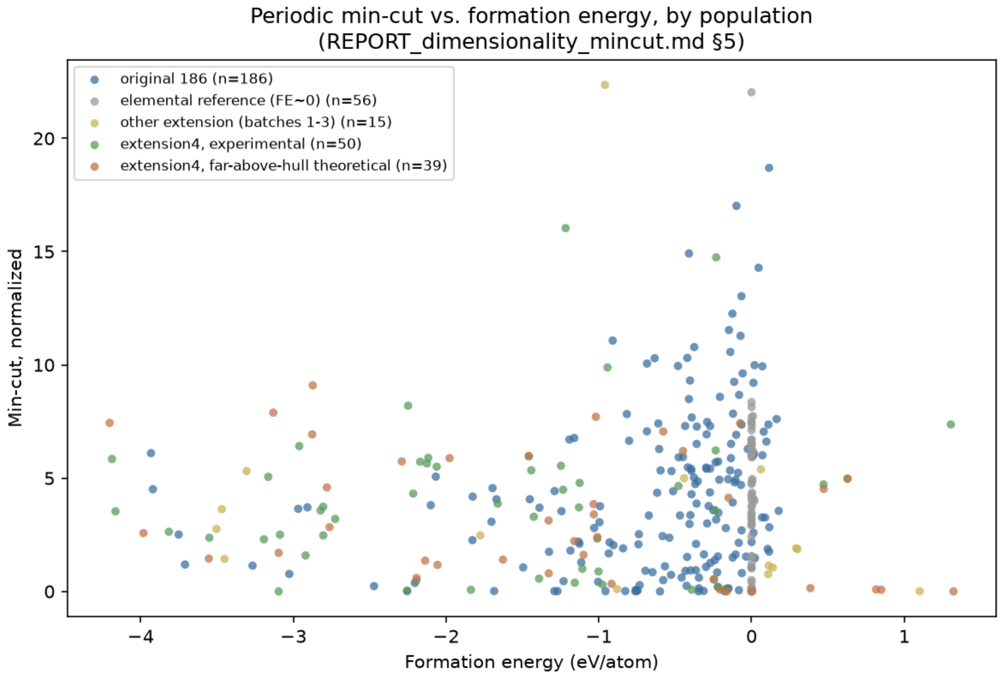
Two new descriptors, `percolation_path.py` left unchanged: `network_dimensionality.py` (0D–3D classification via a relative bond-strength threshold + breadth-first search) and `periodic_mincut.py` (minimum cut via `networkx` max-flow, separability rather than traversability). Headline: the normalized min-cut is the **first descriptor in the project to reach a significant global correlation** ($\rho = 0.285$, $p = 1.0 \times 10^{-4}$, $n = 186$), likely driven in part by between-bond-type clustering; dimensionality alone does not separate `formation_energy_per_atom` (Kruskal–Wallis $p = 0.19$).



Why did the min-cut correlation collapse? Mission #3's headline, $\rho = 0.285$ at $n = 186$, collapsed to $\rho = 0.089$, $p = 0.098$ (not significant) at $n = 349$. **This is not a refutation of the original result** — isolated, the original 186 compounds still give $\rho = 0.2845$, $p = 8.3 \times 10^{-5}$, essentially identical to the mission-#3 number. The collapse is a dilution artifact from pooling populations added for unrelated reasons:

- 65 of the 74 extension-batch-1–3 compounds are single-element reference structures (added for mission #5's elemental references), sitting at `formation_energy_per_atom`≈ 0 by construction with widely-scattered min-cut values — mechanically dilutes a rank correlation regardless of any true relationship. Excluding them recovers $\rho = 0.2157$ ($n = 193$).

- `extension4` is not one population but two, with *opposite-sign* correlations: the experimental half is flat ($\rho = 0.06$, not significant, $n = 50$); the deliberately far-above-hull theoretical half shows a *significant* correlation of the *opposite* sign ($\rho = -0.40$, $p = 0.011$, $n = 39$) — a higher min-cut there is associated with a *more* negative (more stable) formation energy, the reverse of the original population's trend.



This opposite sign is itself resolved (August-16 follow-up, four independent checks on the 39 `theo_far_hull` compounds, `analysis/investigate_theo_far_hull_signflip.py`): neither a cell-size/symmetry selection artifact (survives controlling for `n_sites`+`sg_number`, $\rho = -0.39$) nor a real physical effect of far-from-hull structures, but a **between-anion-chemistry confound** — F-containing systems (high min-cut, strongly negative formation energy) and azide N_3-containing systems (near-zero min-cut, near-zero/positive formation energy) pull the pooled correlation in opposite directions. Controlling for anion identity (dummy regression, $n = 39$) or comparing the experimental/theoretical pair within the exact same chemical system ($n = 37$ pairs) both independently collapse the correlation to $\rho \approx -0.07$, not significant — the same between-group clustering failure mode this project has now hit three times (min-cut/`bond_type`, antibonding/`is_metal`, and now min-cut/anion).

Recommendation: any future min-cut-vs-formation-energy number (or, by the same logic, any other descriptor's) should be conditioned on `family`/selection-provenance, the same way this project already conditions on `bond_type`/`is_metal`.

A.6 Limits specific to the percolation weight

- **Primitive vs. conventional cell**: not reprocessed on the 186-compound dataset. Most likely explanation for the weak observed correlations — the strongest bond can sit entirely inside the primitive cell and contribute to no non-contractile cycle at all, biasing the measured "weakest link." Tested on the 6 pilot compounds (mission #2): the choice changes the percolation weight substantially (3x-55x) but in the *opposite* direction from the hypothesis (the weight gets *smaller* with a larger cell); not extended to the full dataset based on this result.

- **Chance-level reference logistic regression** (AUC ≈ 0.5) with percolation weight as a feature.

- **Why not SISSO**: a single candidate descriptor (the percolation weight, in two algebraically related variants), compared against aggregates that are themselves functions of the same underlying ICOHP values — no meaningful combinatorial search space at this stage.

A.7 Conclusion of the original campaign

The 186-compound dataset was built, computed (VASP+LOBSTER), and analyzed end to end. The headline result is a *negative* but quantified and honest one: at this sample size, the percolation descriptor shows no significant correlation with thermodynamic stability, and demonstrates no advantage over classic ICOHP aggregates. This result, together with the min-cut result (mission #3), motivated the project's reframing around Δ(ICOHP) documented in §6.4.

A.8 Compound List (percolation weight)

Campaign 2's 186 compounds, grouped by category (family), with their Materials Project identifier, PBE thermodynamic hull distance (`energy_above_hull`), and the minimum ICOHP percolation weight (`icohp_percolation_weight_min`). Auto-generated from `analysis/percolation_vs_hull.csv` by `report/gen_appendix.py`.

Table S1: Complete list of the 186 compounds, grouped by category, with their Materials Project identifier, PBE hull distance, and minimum ICOHP percolation weight.

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
Experimental, stable (hull) ($n = 63$)				
As2S3	mp-1189572	covalent	0.0000	1.00e-05
AsF3	mp-28027	covalent	0.0000	0.00e+00
BI3	mp-23189	covalent	0.0000	1.00e-05
BW	mp-7832	–	0.0000	1.90e-04
BaAu2	mp-30363	metallic	0.0000	5.19e-02
BaCl2	mp-568662	ionic	0.0000	1.31e-02
BaH2	mp-23715	–	0.0000	2.40e-04
Be2Cu	mp-2031	metallic	0.0000	1.15e-03
Be2Nb3	mp-11272	metallic	0.0000	6.36e-03
Be2V	mp-11281	metallic	0.0000	6.80e-04
CaAl4	mp-1749	metallic	0.0000	1.03e-02
CaSn	mp-2450	metallic	0.0000	5.23e-03
CdPd	mp-1696	metallic	0.0000	9.00e-05
CoB	mp-20857	–	0.0000	6.00e-05
Cs2As3	mp-15557	–	0.0000	2.02e-02
FeS	mp-505531	–	0.0000	3.05e-03
Ge2Os	mp-10032	metallic	0.0000	1.21e-02
HfTc2	mp-1095669	metallic	0.0000	1.77e-02
Hg4Pt	mp-936	metallic	0.0000	3.36e-03
HgF2	mp-8177	–	0.0000	8.95e-03
InBr	mp-22870	covalent	0.0000	1.02e-02
KI	mp-22898	ionic	0.0000	2.26e-02
KN3	mp-827	–	0.0000	2.80e-04
Li2Se	mp-2286	–	0.0000	1.16e-02
LiB	mp-1001835	–	0.0000	2.91e-03
Mg2Ni	mp-2137	metallic	0.0000	1.60e-04
MgP4	mp-384	–	0.0000	0.00e+00
MnAl12	mp-1104165	metallic	0.0000	1.42e-02
MoSe2	mp-1634	covalent	0.0000	1.16e-03
NaS	mp-2400	–	0.0000	0.00e+00
Na	mp-10172	–	0.0000	1.97e-03
Nb3Ir	mp-1458	metallic	0.0000	9.02e-03
NbGa3	mp-1973	metallic	0.0000	8.58e-03
PI2	mp-29443	covalent	0.0000	2.30e-04
PbSe	mp-2201	covalent	0.0000	1.66e-02

Table S1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
RbI	mp-22903	ionic	0.0000	2.00e-02
Sc5Ge3	mp-17190	metallic	0.0000	5.30e-04
ScAg2	mp-1018128	metallic	0.0000	1.70e-04
ScTe	mp-10026	–	0.0000	7.12e-03
SnPt	mp-19856	metallic	0.0000	1.29e-02
SrO2	mp-2697	ionic	0.0000	4.40e-04
SrSi	mp-2661	metallic	0.0000	7.00e-05
Ta5Ge3	mp-17593	metallic	0.0000	3.46e-02
TaP	mp-1067587	–	0.0000	3.14e-02
TaSb2	mp-11697	metallic	0.0000	1.17e-02
Te2Pd	mp-782	–	0.0000	1.44e-02
Te2Ru	mp-267	covalent	0.0000	4.34e-03
TeO2	mp-8377	covalent	0.0000	4.00e-05
Ti3In	mp-866046	metallic	0.0000	5.01e-03
Ti3Pt	mp-2339	metallic	0.0000	1.78e-03
TiBe12	mp-1104067	metallic	0.0000	2.40e-04
TiNi	mp-1048	metallic	0.0000	3.91e-03
TiO	mp-1071163	–	0.0000	6.40e-04
TiSi2	mp-1077503	metallic	0.0000	6.18e-03
TlCl	mp-571079	–	0.0000	7.17e-03
VNi2	mp-11531	metallic	0.0000	3.88e-03
WO2	mp-1524394	–	0.0000	2.20e-04
Zr2Au	mp-2348	metallic	0.0000	3.06e-02
Zr5Pb3	mp-681992	metallic	0.0000	2.00e-05
ZrMo2	mp-2049	metallic	0.0000	1.05e-02
Si	mp-149	covalent	0.0000	3.17e-03
NaCl	mp-22862	ionic	0.0000	3.07e-02
AlNi	mp-1487	metallic	0.0000	4.19e-03
Experimental, metastable ($n = 63$)				
CdI2	mp-567259	–	0.0005	2.38e-02
MgCl2	mp-570259	ionic	0.0006	1.90e-04
C	mp-169	covalent	0.0008	2.50e-04
ZrBr	mp-1065846	–	0.0008	7.88e-03
Li3Hg	mp-1646	metallic	0.0013	1.20e-03
Ta7Co6	mp-1104548	metallic	0.0015	3.10e-04
CaGa4	mp-2461	metallic	0.0016	1.91e-02
InCl	mp-571636	covalent	0.0016	8.50e-03
NiB	mp-14019	–	0.0018	4.10e-04
MgH2	mp-23711	–	0.0019	1.00e-05
Sn7Ir5	mp-20466	metallic	0.0024	1.00e-02
NaN3	mp-1066400	–	0.0029	7.00e-04
ClF3	mp-556767	–	0.0030	0.00e+00
SiS2	mp-1095294	covalent	0.0030	1.00e-04
Sr5Sn3	mp-17720	metallic	0.0030	1.08e-02
Te2Ir	mp-1551	–	0.0030	2.80e-03
Li13Sn5	mp-30769	metallic	0.0033	8.50e-04
H3N	mp-643432	covalent	0.0038	0.00e+00
Cs2Se	mp-1011696	–	0.0042	2.62e-02
In3Ir	mp-630976	metallic	0.0043	2.04e-03
HI	mp-697025	–	0.0044	0.00e+00
Rb2Hg7	mp-31474	metallic	0.0045	3.14e-02
PbAu2	mp-19871	metallic	0.0047	2.01e-03
BeCu	mp-2323	metallic	0.0050	4.80e-04
SiO2	mp-546794	covalent	0.0056	1.50e-04

Table S1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
IF7	mp-27988	–	0.0063	0.00e+00
K	mp-10157	–	0.0069	8.07e-02
Al3Os2	mp-16521	metallic	0.0074	6.23e-03
Nb2O5	mp-1595	–	0.0075	1.00e-04
LiBr	mp-23259	ionic	0.0099	9.88e-02
ZnAg	mp-1912	metallic	0.0112	9.67e-03
MgPd3	mp-7753	metallic	0.0114	2.21e-03
AsPd2	mp-1080831	–	0.0118	3.50e-04
GeSe2	mp-10074	covalent	0.0125	4.00e-05
Re3Ge7	mp-29141	metallic	0.0126	2.42e-03
Cu3Ge	mp-19724	metallic	0.0126	5.30e-04
Ti3Ge	mp-1189044	metallic	0.0130	1.40e-02
ZrGa3	mp-30685	metallic	0.0130	7.64e-03
SrAl2	mp-1638	metallic	0.0136	5.36e-03
NiS	mp-1547	–	0.0142	3.00e-04
Al12W	mp-11227	metallic	0.0152	1.37e-02
CsH	mp-632319	–	0.0161	2.66e-02
CaGa2	mp-992	metallic	0.0170	1.66e-02
NiHg4	mp-570835	metallic	0.0171	1.09e-03
SiRh	mp-2287	metallic	0.0246	1.10e-02
BaSn5	mp-571353	metallic	0.0262	4.58e-03
PS	mp-612	covalent	0.0265	8.00e-05
AsRh	mp-22079	–	0.0269	1.93e-03
Te2Au	mp-567525	–	0.0274	1.62e-02
Be3N2	mp-6977	–	0.0315	2.00e-05
Y2O3	mp-558573	–	0.0342	5.00e-05
AgO	mp-1079720	–	0.0423	4.00e-05
FeTe2	mp-1102002	–	0.0449	5.20e-04
H2O	mp-2739191	–	0.0494	0.00e+00
CdO2	mp-2310	–	0.0562	9.00e-05
Mo3Se4	mp-21021	–	0.0609	1.84e-03
PbSe	mp-22009	covalent	0.0731	1.68e-02
Be12Pd	mp-12666	metallic	0.0795	5.00e-05
Sr2As	mp-15697	–	0.0800	2.06e-03
AsRh2	mp-1102364	–	0.0806	4.63e-03
ZrRh	mp-2808	metallic	0.0839	3.50e-02
K2S	mp-559278	–	0.0873	1.07e-02
Ge4Rh	mp-1104286	metallic	0.0907	2.08e-03
Theoretical, metastable ($n = 60$)				
Li3Ag	mp-976408	metallic	0.0013	1.15e-02
Y3Co	mp-1189329	metallic	0.0016	1.50e-02
TiNi	mp-1067248	metallic	0.0020	5.29e-03
TaP	mp-1187244	–	0.0033	3.59e-02
Sr3As4	mp-678007	–	0.0062	4.47e-03
Ga2Se3	mp-1224809	covalent	0.0063	1.14e-02
MnN	mp-1009130	covalent	0.0151	1.85e-03
BeCl2	mp-1190080	ionic	0.0187	2.37e-03
MgSn	mp-1094801	metallic	0.0212	4.26e-03
TiW	mp-1216621	metallic	0.0225	2.25e-03
SrTl3	mp-1206636	metallic	0.0294	3.29e-03
IrCl3	mp-729507	–	0.0301	2.72e-03
Ta4C3	mp-1218000	–	0.0335	1.44e-03
Sc2O3	mp-755313	–	0.0364	5.10e-04
SnBr2	mp-978985	covalent	0.0405	3.40e-04

Table S1 (continued)

Formula	mp-id	Bonding	E_{hull} (eV/at)	Percolation weight (eV)
Sn7Pd5	mp-1219006	metallic	0.0414	2.30e-03
Tl4Sn	mp-1216560	metallic	0.0459	1.00e+00
LiTl3	mp-1185253	metallic	0.0467	3.35e-03
PtCl2	mp-1219748	–	0.0490	5.04e-03
HCl	mp-684609	–	0.0536	2.50e-04
MgSn	mp-1094669	metallic	0.0560	2.60e-04
K5As4	mp-684799	–	0.0564	8.80e-03
KSn3	mp-1185151	metallic	0.0573	2.65e-02
NbS2	mp-774873	–	0.0576	1.20e-04
YAg3	mp-1187811	metallic	0.0594	1.12e-03
B3Ir2	mp-1228647	–	0.0609	2.80e-04
TiMo2	mp-1216675	metallic	0.0613	3.77e-02
Cr2C	mp-1226378	–	0.0625	2.08e-03
ZrO2	mp-775909	–	0.0627	1.20e-04
Zn	mp-2647117	–	0.0717	1.04e-02
IN2	mp-1097011	–	0.0730	4.70e-04
CoSe2	mp-1390196	–	0.0734	8.40e-04
GeS	mp-12910	covalent	0.0762	2.26e-03
ZrTi	mp-1215236	metallic	0.0883	3.15e-02
VCo3	mp-1095057	metallic	0.0942	2.93e-03
Tl3Ga	mp-1187539	metallic	0.0972	5.00e-05
CaGe2	mp-643542	metallic	0.0979	1.30e-02
NbRh2	mp-1220414	metallic	0.1043	1.76e-02
TcW	mp-1217337	metallic	0.1056	4.10e-03
MgF2	mp-560236	ionic	0.1076	2.00e-05
VN	mp-1017532	–	0.1080	4.00e-05
Na2Cl	mp-990084	–	0.1105	3.76e-02
ZnPb3	mp-1187949	metallic	0.1107	4.51e-03
Sr3Sn	mp-1187136	metallic	0.1111	7.94e-03
TaRe	mp-1217894	metallic	0.1113	6.98e-02
Al4Si	mp-1228120	metallic	0.1149	8.54e-02
B6Pb	mp-1096825	covalent	0.1151	2.35e-03
ScTc3	mp-867262	metallic	0.1154	3.31e-02
CrMo	mp-1226200	metallic	0.1163	4.23e-02
BPd5	mp-1228665	–	0.1325	2.42e-02
Zr3Be	mp-1188016	metallic	0.1430	1.05e-03
Na3Cl	mp-1064484	–	0.1456	2.55e-03
SiAu3	mp-1186998	metallic	0.1643	5.68e-03
InSe2	mp-1232036	–	0.1733	1.64e-03
TiSi2	mp-570991	metallic	0.1796	1.35e-02
BeAu3	mp-983539	metallic	0.1818	2.41e-02
Zr3Ge	mp-1188039	metallic	0.1851	6.59e-03
AlSb	mp-1023918	metallic	0.1878	3.60e-04
Mg15B	mp-1023515	–	0.1943	2.80e-04
HfMo	mp-1224314	metallic	0.1991	2.00e-02

B VASP Calculation Parameters

Fixed INCAR parameters, identical across the whole dataset (static, LOBSTER-compatible run) — only `NBANDS` (appendix D) and the k-point grid vary by compound.

Parameter	Value	Description
PREC	Accurate	Overall precision (charge grid, projectors)
ENCUT	600.0	Plane-wave cutoff energy (eV)
EDIFF	1e-06	Electronic convergence criterion (eV)
IBRION	-1	Ionic dynamics type (−1 = static, no relaxation)
NSW	0	Number of ionic steps (0 = static run on the MP structure)
ISMEAR	0	Smearing type (0 = Gaussian)
SIGMA	0.05	Smearing width (eV)
LASPH	True	Non-spherical corrections to the PAW potential
ISYM	-1	Symmetry (−1 = disabled, required by LOBSTER)
LWAVE	True	Write WAVECAR (required by LOBSTER)
LCHARG	True	Write CHGCAR
NPAR	4	Band-level parallelization
KPAR	2	K-point-level parallelization

C PAW-PBE Potentials

One PAW-PBE potential per element present in the dataset (pymatgen's `MPRelaxSet` library — the same one Materials Project used to relax the input structures), with one documented exception for tungsten (§7): the recommended `W_pv` variant is absent from the local mirror, and `W_sv` turned out to be corrupted (missing `LEXCH` field) — plain `W` is used instead, verified valid.

Table S2: PAW-PBE potentials used, one per element present in the dataset (pymatgen's MPRelaxSet library, with one documented exception: W substituted for W_pv/W_sv, see §7).

Element	PAW-PBE potential
Ag	Ag (02Apr2005)
Al	Al (04Jan2001)
As	As (22Sep2009)
Au	Au (04Oct2007)
B	B (06Sep2000)
Ba	Ba_sv (06Sep2000)
Be	Be_sv (06Sep2000)
Br	Br (06Sep2000)
C	C (08Apr2002)
Ca	Ca_sv (06Sep2000)
Cd	Cd (06Sep2000)
Cl	Cl (06Sep2000)
Co	Co (02Aug2007)
Cr	Cr_pv (02Aug2007)
Cs	Cs_sv (25Jan2019)
Cu	Cu_pv (06Sep2000)
F	F (08Apr2002)
Fe	Fe_pv (02Aug2007)
Ga	Ga_d (06Jul2010)
Ge	Ge_d (03Jul2007)
H	H (15Jun2001)
Hf	Hf_pv (06Sep2000)

Element	PAW-PBE potential
Hg	`Hg (06Sep2000)`
I	`I (08Apr2002)`
In	`In_d (06Sep2000)`
Ir	`Ir (06Sep2000)`
K	`K_sv (06Sep2000)`
Li	`Li_sv (10Sep2004)`
Mg	`Mg_pv (13Apr2007)`
Mn	`Mn_pv (02Aug2007)`
Mo	`Mo_pv (04Feb2005)`
N	`N (08Apr2002)`
Na	`Na_pv (19Sep2006)`
Nb	`Nb_pv (08Apr2002)`
Ni	`Ni_pv (06Sep2000)`
O	`O (08Apr2002)`
Os	`Os_pv (20Jan2003)`
P	`P (06Sep2000)`
Pb	`Pb_d (06Sep2000)`
Pd	`Pd (04Jan2005)`
Pt	`Pt (04Feb2005)`
Rb	`Rb_sv (06Sep2000)`
Re	`Re_pv (06Sep2000)`
Rh	`Rh_pv (25Jan2005)`
Ru	`Ru_pv (28Jan2005)`
S	`S (06Sep2000)`
Sb	`Sb (06Sep2000)`
Sc	`Sc_sv (07Sep2000)`
Se	`Se (06Sep2000)`
Si	`Si (05Jan2001)`
Sn	`Sn_d (06Sep2000)`
Sr	`Sr_sv (07Sep2000)`
Ta	`Ta_pv (07Sep2000)`
Tc	`Tc_pv (04Feb2005)`
Te	`Te (08Apr2002)`
Ti	`Ti_pv (07Sep2000)`
Tl	`Tl_d (06Sep2000)`
V	`V_pv (07Sep2000)`
W	`W (08Apr2002)`
Y	`Y_sv (25May2007)`
Zn	`Zn (06Sep2000)`
Zr	`Zr_sv (04Jan2005)`

D k-point Grid and NBANDS per Compound

k-point grid auto-generated by `pymatgen.io.vasp.Kpoints.automatic_density_by_vol` (density `kppvol=100`, Γ-centered or Monkhorst–Pack mesh depending on cell symmetry) and `NBANDS`, derived per compound from the number of local orbitals in the LOBSTER basis (`Lobsterin.write_INCAR`) — a strictly sufficient value for the local-basis projection, not a fixed guess.

Table S3: k-point grid (`pymatgen` automatic density, `kppvol=100`) and number of bands (derived from the LOBSTER basis) per compound.

Formula	mp-id	k-point grid	NBANDS
Experimental, stable (hull) ($n = 63$)			
As2S3	mp-1189572	5 3 2 (G)	80
AsF3	mp-28027	5 4 4 (G)	64
BI3	mp-23189	4 4 4 (G)	32
BW	mp-7832	9 9 3 (G)	52
BaAu2	mp-30363	6 6 7 (G)	23
BaCl2	mp-568662	6 6 6 (G)	13
BaH2	mp-23715	6 4 3 (G)	28
Be2Cu	mp-2031	7 7 7 (G)	38
Be2Nb3	mp-11272	8 4 4 (M)	74
Be2V	mp-11281	7 7 4 (G)	76
CaAl4	mp-1749	7 7 4 (G)	21
CaSn	mp-2450	6 6 4 (M)	28
CdPd	mp-1696	7 6 6 (G)	36
CoB	mp-20857	9 7 5 (G)	52
Cs2As3	mp-15557	4 4 3 (G)	44
FeS	mp-505531	8 8 5 (G)	26
Ge2Os	mp-10032	9 6 4 (G)	54
HfTc2	mp-1095669	5 5 3 (G)	108
Hg4Pt	mp-936	5 5 5 (G)	45
HgF2	mp-8177	8 8 8 (G)	17
InBr	mp-22870	6 6 4 (M)	26
KI	mp-22898	6 6 6 (G)	9
KN3	mp-827	5 5 5 (G)	34
Li2Se	mp-2286	7 7 7 (G)	14
LiB	mp-1001835	7 7 9 (G)	18
Mg2Ni	mp-2137	5 5 2 (G)	102
MgP4	mp-384	5 5 3 (G)	40
MnAl12	mp-1104165	4 4 4 (M)	57
MoSe2	mp-1634	9 9 2 (G)	34
NaS	mp-2400	6 6 3 (G)	32
Na	mp-10172	8 8 4 (G)	8
Nb3Ir	mp-1458	5 5 5 (G)	72
NbGa3	mp-1973	8 8 6 (M)	36
PI2	mp-29443	6 4 3 (G)	24
PbSe	mp-2201	7 7 7 (G)	13
RbI	mp-22903	6 6 6 (G)	9
Sc5Ge3	mp-17190	3 3 5 (G)	154
ScAg2	mp-1018128	8 8 5 (G)	28
ScTe	mp-10026	7 7 4 (G)	28
SnPt	mp-19856	7 7 5 (G)	36
SrO2	mp-2697	8 8 7 (G)	13
SrSi	mp-2661	7 6 4 (G)	18
Ta5Ge3	mp-17593	4 4 4 (M)	144
TaP	mp-1067587	9 9 4 (G)	26
TaSb2	mp-11697	8 5 4 (G)	34
Te2Pd	mp-782	7 7 5 (G)	17
Te2Ru	mp-267	7 5 4 (G)	34
TeO2	mp-8377	6 5 3 (G)	48
Ti3In	mp-866046	5 5 6 (G)	72
Ti3Pt	mp-2339	5 5 5 (G)	72
TiBe12	mp-1104067	7 5 5 (G)	69

Table S3 (continued)

Formula	mp-id	k-point grid	NBANDS
TiNi	mp-1048	10 7 6 (G)	36
TiO	mp-1071163	10 6 6 (G)	39
TiSi2	mp-1077503	8 8 4 (M)	34
TlCl	mp-571079	6 6 4 (M)	26
VNi2	mp-11531	12 8 7 (G)	27
WO2	mp-1524394	6 5 5 (G)	68
Zr2Au	mp-2348	9 9 4 (G)	29
Zr5Pb3	mp-681992	3 3 5 (G)	154
ZrMo2	mp-2049	6 6 6 (G)	56
Si	mp-149	8 8 8 (G)	100
NaCl	mp-22862	8 8 8 (G)	100
AlNi	mp-1487	10 10 10 (M)	100
Experimental, metastable ($n = 63$)			
CdI2	mp-567259	7 7 4 (G)	17
MgCl2	mp-570259	8 8 5 (G)	12
C	mp-169	12 12 8 (M)	100
ZrBr	mp-1065846	8 8 3 (G)	28
Li3Hg	mp-1646	7 7 7 (G)	24
Ta7Co6	mp-1104548	6 6 3 (G)	117
CaGa4	mp-2461	7 7 5 (G)	41
InCl	mp-571636	6 6 4 (M)	26
NiB	mp-14019	10 10 7 (G)	26
MgH2	mp-23711	6 5 5 (G)	24
Sn7Ir5	mp-20466	4 4 4 (M)	108
NaN3	mp-1066400	7 7 7 (G)	16
ClF3	mp-556767	6 4 3 (G)	64
SiS2	mp-1095294	5 4 3 (G)	48
Sr5Sn3	mp-17720	3 3 3 (G)	104
Te2Ir	mp-1551	7 5 2 (G)	51
Li13Sn5	mp-30769	6 6 1 (G)	110
H3N	mp-643432	8 5 5 (G)	28
Cs2Se	mp-1011696	5 3 2 (G)	56
In3Ir	mp-630976	4 4 4 (M)	144
HI	mp-697025	3 3 3 (G)	40
Rb2Hg7	mp-31474	4 4 4 (G)	73
PbAu2	mp-19871	5 5 5 (G)	54
BeCu	mp-2323	10 10 10 (M)	100
SiO2	mp-546794	6 6 6 (M)	24
IF7	mp-27988	5 5 3 (G)	64
K	mp-10157	6 6 6 (G)	5
Al3Os2	mp-16521	9 9 3 (G)	30
Nb2O5	mp-1595	7 7 4 (G)	38
LiBr	mp-23259	8 8 8 (G)	100
ZnAg	mp-1912	9 9 9 (G)	18
MgPd3	mp-7753	7 7 7 (G)	31
AsPd2	mp-1080831	8 4 4 (G)	66
GeSe2	mp-10074	5 5 5 (G)	34
Re3Ge7	mp-29141	9 6 1 (G)	180
Cu3Ge	mp-19724	6 6 5 (G)	72
Ti3Ge	mp-1189044	4 4 4 (M)	144
ZrGa3	mp-30685	7 7 5 (G)	37
SrAl2	mp-1638	5 5 5 (G)	26
NiS	mp-1547	6 6 6 (M)	39
Al12W	mp-11227	4 4 4 (M)	57

Table S3 (continued)

Formula	mp-id	k-point grid	NBANDS
CsH	mp-632319	7 7 7 (G)	6
CaGa2	mp-992	7 7 6 (G)	23
NiHg4	mp-570835	6 6 6 (M)	45
SiRh	mp-2287	6 6 5 (G)	52
BaSn5	mp-571353	5 5 4 (G)	50
PS	mp-612	4 4 3 (G)	64
AsRh	mp-22079	8 5 4 (G)	52
Te2Au	mp-567525	7 7 5 (G)	17
Be3N2	mp-6977	10 10 3 (G)	46
Y2O3	mp-558573	8 4 3 (G)	96
AgO	mp-1079720	8 5 5 (G)	52
FeTe2	mp-1102002	4 4 4 (M)	68
H2O	mp-2739191	7 7 4 (G)	24
CdO2	mp-2310	5 5 5 (G)	68
Mo3Se4	mp-21021	4 4 4 (M)	86
PbSe	mp-22009	6 6 2 (G)	26
Be12Pd	mp-12666	6 6 6 (M)	69
Sr2As	mp-15697	6 6 3 (G)	28
AsRh2	mp-1102364	7 4 3 (G)	88
ZrRh	mp-2808	8 8 8 (M)	19
K2S	mp-559278	5 5 4 (G)	28
Ge4Rh	mp-1104286	4 4 3 (G)	135
Theoretical, metastable ($n = 60$)			
Li3Ag	mp-976408	7 7 7 (G)	24
Y3Co	mp-1189329	3 4 4 (G)	156
TiNi	mp-1067248	10 7 6 (G)	36
TaP	mp-1187244	9 9 9 (G)	13
Sr3As4	mp-678007	3 3 4 (G)	62
Ga2Se3	mp-1224809	5 5 5 (G)	30
MnN	mp-1009130	10 10 10 (G)	13
BeCl2	mp-1190080	3 3 3 (G)	78
MgSn	mp-1094801	8 8 6 (M)	13
TiW	mp-1216621	10 10 6 (M)	18
SrTl3	mp-1206636	5 5 5 (G)	32
IrCl3	mp-729507	3 3 6 (G)	84
Ta4C3	mp-1218000	6 6 6 (M)	48
Sc2O3	mp-755313	6 5 6 (G)	64
SnBr2	mp-978985	4 4 6 (M)	34
Sn7Pd5	mp-1219006	5 5 3 (G)	108
Tl4Sn	mp-1216560	5 5 5 (G)	45
LiTl3	mp-1185253	6 6 6 (M)	32
PtCl2	mp-1219748	5 5 4 (G)	34
HCl	mp-684609	7 7 7 (G)	5
MgSn	mp-1094669	8 8 8 (M)	13
K5As4	mp-684799	4 4 3 (G)	41
KSn3	mp-1185151	4 4 5 (G)	64
NbS2	mp-774873	10 6 2 (M)	34
YAg3	mp-1187811	6 6 6 (G)	37
B3Ir2	mp-1228647	9 5 4 (G)	60
TiMo2	mp-1216675	5 6 14 (G)	27
Cr2C	mp-1226378	10 10 7 (G)	22
ZrO2	mp-775909	10 6 4 (M)	36
Zn	mp-2647117	11 11 11 (G)	9
IN2	mp-1097011	5 5 5 (G)	24

Table S3 (continued)

Formula	mp-id	k-point grid	NBANDS
CoSe2	mp-1390196	4 4 4 (G)	68
GeS	mp-12910	7 7 5 (G)	26
ZrTi	mp-1215236	9 9 6 (G)	19
VCo3	mp-1095057	7 6 6 (G)	72
Tl3Ga	mp-1187539	5 5 5 (G)	36
CaGe2	mp-643542	7 7 6 (G)	23
NbRh2	mp-1220414	10 6 4 (M)	54
TcW	mp-1217337	10 10 6 (M)	18
MgF2	mp-560236	6 6 5 (G)	24
VN	mp-1017532	11 11 5 (G)	26
Na2Cl	mp-990084	6 6 6 (M)	12
ZnPb3	mp-1187949	5 5 5 (G)	36
Sr3Sn	mp-1187136	5 5 5 (G)	24
TaRe	mp-1217894	10 10 6 (M)	18
Al4Si	mp-1228120	6 6 6 (M)	20
B6Pb	mp-1096825	6 6 6 (M)	33
ScTc3	mp-867262	7 7 7 (G)	37
CrMo	mp-1226200	11 11 7 (G)	18
BPd5	mp-1228665	6 6 6 (M)	49
Zr3Be	mp-1188016	6 6 6 (M)	35
Na3Cl	mp-1064484	8 8 3 (G)	16
SiAu3	mp-1186998	7 7 7 (G)	31
InSe2	mp-1232036	7 7 3 (G)	34
TiSi2	mp-570991	8 8 8 (M)	17
BeAu3	mp-983539	7 7 7 (G)	32
Zr3Ge	mp-1188039	4 4 6 (G)	78
AlSb	mp-1023918	6 6 5 (G)	16
Mg15B	mp-1023515	4 4 3 (G)	64
HfMo	mp-1224314	10 10 6 (M)	18

E Elements and Allotropes

Every single-element structure with no matched polymorph in the dataset: the elemental references (`ELEMENT_REFERENCE`) used to decompose multi-element compounds. An experimental compound's formula is marked with a superscript star (black if it is the stable phase, red if it is a metastable polymorph/allotrope); unmarked formulas are theoretical. Auto-generated by `report/gen_appendix_extension.py`.

Table S4: Elements and allotropes with no matched polymorph in the `extension` campaign: references used to decompose multi-element compounds. Allotropes with one or more other allotropes computed in this dataset (e.g. carbon) are grouped in Table S5 instead. Black * = stable experimental reference (thermodynamic phase); red * = metastable experimental reference; unmarked = theoretical. No ΔICOHP column: an element is not itself a decomposition reaction.

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
Ag*	mp-124	Fm-3m	0.0021
Al*	mp-134	Fm-3m	0.0000
As*	mp-158	Cmce	0.0000
Au*	mp-81	Fm-3m	0.0000

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
B*	mp-160	R-3m	0.0000
Ba*	mp-122	Im-3m	0.0000
Be*	mp-87	P6_3/mmc	0.0000
Br*	mp-23154	Cmce	0.0000
Ca*	mp-21	Im-3m	0.0056
Cd*	mp-94	P6_3/mmc	0.0000
Cl_2*	mp-22848	Cmce	0.0000
Co*	mp-102	Fm-3m	0.0000
Cr*	mp-90	Im-3m	0.0000
Cs*	mp-1	Im-3m	0.0193
Cu*	mp-30	Fm-3m	0.0000
F_2*	mp-561203	C2/c	0.0031
Fe*	mp-13	Im-3m	0.0000
Ga*	mp-142	Cmce	0.0000
Ge*	mp-32	Fd-3m	0.0000
H_2*	mp-730101	P2_12_12_1	0.0000
Hf*	mp-103	P6_3/mmc	0.0000
Hg*	mp-10861	P6/mmm	0.0030
I*	mp-23153	Cmce	0.0000
In*	mp-1055994	I4/mmm	0.0045
Ir*	mp-101	Fm-3m	0.0000
Li*	mp-1018134	R-3m	0.0000
Mg*	mp-153	P6_3/mmc	0.0000
Mn*	mp-35	I-43m	0.0000
Mo*	mp-129	Im-3m	0.0000
N_2	mp-1059834	Imma	1.1010
Nb*	mp-75	Im-3m	0.0000
Ni*	mp-23	Fm-3m	0.0000
O_2*	mp-1524462	C2/m	0.0000
Os*	mp-49	P6_3/mmc	0.0000
P*	mp-568348	P2/c	0.0000
Pb*	mp-20483	Fm-3m	0.0000
Pd*	mp-2	Fm-3m	0.0000
Pt*	mp-126	Fm-3m	0.0000
Rb*	mp-70	Im-3m	0.0092
Re*	mp-8	P6_3/mmc	0.0036
Rh*	mp-74	Fm-3m	0.0000
Ru*	mp-33	P6_3/mmc	0.0000
S*	mp-77	Fddd	0.0000
Sb*	mp-104	R-3m	0.0000
Sc*	mp-67	P6_3/mmc	0.0000
Se*	mp-14	P3_121	0.0011
Sn*	mp-623511	I4/mmm	0.0614
Sr*	mp-139	P6_3/mmc	0.0000
Ta*	mp-50	Im-3m	0.0091
Tc*	mp-113	P6_3/mmc	0.0000
Te*	mp-19	P3_121	0.0000
Ti*	mp-46	P6_3/mmc	0.0152
Tl*	mp-82	P6_3/mmc	0.0000
V*	mp-146	Im-3m	0.0000
W*	mp-91	Im-3m	0.0000
Y*	mp-112	P6_3/mmc	0.0000
Zn*	mp-79	P6_3/mmc	0.0000
Zr*	mp-131	P6_3/mmc	0.0000

F Polymorphs and Allotropes

Elements and compounds with several entries sharing a formula in the dataset, grouped here rather than scattered across the elements/compounds tables (see the table caption for detail).

Table S5: Elements (allotropes) and compounds (polymorphs) with **several entries sharing a formula** in the `extension` campaign (39 groups, 84 entries) — grouped here rather than scattered across the elements/compounds tables, since the natural comparison for a polymorph group is polymorph-to-polymorph, not the decomposition-to-elements reaction (hence no ΔICOHP column). A rule separates each group; within a group, sorted by increasing E_{hull}. Black * = stable experimental; red * = metastable experimental; unmarked = theoretical.

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
BaF_3*	mp-1080821	P2_1/m	0.0000
BaF_3	mp-1183303	I4/mmm	0.0612
BaO*	mp-1008500	P6_3/mmc	0.0189
BaO	mp-1950989	P6_3/mmc	0.0434
Be_3N_2*	mp-18337	Ia-3	0.0000
Be_3N_2	mp-1017550	P-3m1	0.2081
Be_3P_2*	mp-567841	Ia-3	0.0000
Be_3P_2	mp-1084771	Pn-3m	0.1567
$BeCl_2$*	mp-23267	Ibam	0.0087
$BeCl_2$	mp-1172909	I-43m	3.1688
BeF_2*	mp-15951	P3_121	0.0004
BeF_2	mp-975644	P6/mmm	2.5181
BeO	mp-1778	F-43m	0.0070
BeO*	mp-7599	P4_2/mnm	0.0153
C*	mp-48	P6_3/mmc	0.0031
C*	mp-66	Fd-3m	0.1123
C*	mp-47	P6_3/mmc	0.1395
C	mp-1190171	Pnma	0.2927
C	mp-1080826	C2/m	0.3009
Ca_3N_2*	mp-844	Ia-3	0.0000
Ca_3N_2*	mp-1047	R-3c	0.0167
Ca_3N_2	mp-1013524	Pm-3m	0.5112
$CaCl_2$*	mp-23214	Pnnm	0.0011
$CaCl_2$	mp-1120739	Fm-3m	0.0422
CaF_2*	mp-10464	Pnma	0.0346
CaF_2	mp-554355	P4/mmm	0.2403
CaO*	mp-2605	Fm-3m	0.0000
CaO*	mp-1079707	Cmce	0.2098
CaO	mp-1181903	Fd-3m	2.7281
CsN_3*	mp-510557	I4/mcm	0.0000
CsN_3	mp-1225892	P4/mmm	0.0113
CsO_2*	mp-1441	I4/mmm	0.0349
CsO_2	mp-1096936	Cmmm	0.0940
CsP_7*	mp-1201790	Pbcm	0.0000

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
CsP_7	mp-1213206	Pca2_1	0.0001
K_2O^*	mp-971	Fm-3m	0.0000
K_2O	mp-1223784	P4/mmm	0.2000
KF_3	mp-2749823	P2_1	0.0119
$KF_3{}^*$	mp-1544557	P2_12_12_1	0.0125
KP^*	mp-1058315	R-3m	1.0080
KP	mp-1057787	Immm	1.0092
Li_2O^*	mp-1960	Fm-3m	0.0000
Li_2O	mp-755894	Pnma	0.0844
Li_2S^*	mp-1125	Pnma	0.0615
Li_2S	mp-557142	P6_3/mmc	0.1750
LiCl	mp-1185319	P6_3mc	0.0039
$LiCl^*$	mp-22905	Fm-3m	0.0243
LiF^*	mp-1138	Fm-3m	0.0000
LiF	mp-1009009	Pm-3m	0.2875
$LiP_5{}^*$	mp-2412	Pna2_1	0.0077
LiP_5	mp-32760	Pna2_1	0.0963
$MgCl_2{}^*$	mp-23210	R-3m	0.0000
$MgCl_2$	mp-569626	Ama2	0.1991
$MgF_2{}^*$	mp-1072956	Pnnm	0.0025
MgF_2	mp-1062842	Amm2	0.2636
MgO^*	mp-549706	P6_3mc	0.1180
MgO	mp-1009127	Pm-3m	0.7484
MgS^*	mp-1315	Fm-3m	0.0000
MgS	mp-13032	F-43m	0.0343
Na_2O^*	mp-2352	Fm-3m	0.0000
Na_2O	mp-1975297	C2/m	0.0981
$NaN_3{}^*$	mp-22003	R-3m	0.0017
NaN_3	mp-1065265	C2/m	1.0162
NaS^*	mp-409	P-62m	0.0023
NaS	mp-1180106	C2/m	1.0061
Rb_2O^*	mp-1394	Fm-3m	0.0032
Rb_2O	mp-2023482	P-3m1	0.0479
RbF	mp-11718	Fm-3m	0.0000
RbF^*	mp-2064	Pm-3m	0.0516
$RbN_3{}^*$	mp-581833	P4/mmm	0.0193
RbN_3	mp-1219567	Amm2	0.7966
RbP^*	mp-1179688	C2/m	0.9879
RbP	mp-1058499	Immm	0.9882
RbS^*	mp-558071	Immm	0.0019
RbS	mp-1179684	C2/m	0.9975
$Sr_3P_4{}^*$	mp-14288	Fdd2	0.0000
Sr_3P_4	mp-682953	P1	0.0081
SrF_2	mp-1102911	Pnma	0.0267
$SrF_2{}^*$	mp-1102240	Pnma	0.0646
$SrO_2{}^*$	mp-726705	Pnma	0.0096

Formula	mp-id / COD	Space group	E_{hull} (eV/at)
SrO_2	mp-2763179	Pnma	0.0110
TiO_2*	mp-1439	Pbcn	0.0045
TiO_2*	mp-430	P2_1/c	0.0395
TiO_2*	mp-9173	Pnma	0.0575

G Target Compounds

Every multi-element target compound with no matched polymorph in the dataset, with ΔICOHP per atom and its endobondic/exobondic label where a case-1 decomposition reaction could be computed. Auto-generated from `analysis/reaction_icohp_case1.csv` and `analysis/reaction_analysis_ca` by `report/gen_appendix_extension.py`.

Table S6: Target compounds with no matched polymorph in the `extension` campaign. Formulas with several entries in this dataset are grouped in Table S5 instead. Black * = stable experimental compound (thermodynamic phase); red * = metastable experimental compound; unmarked = theoretical. ΔICOHP per atom, products − reactants convention (`reaction_analysis`); "–" where no case-1 reaction could be computed (missing elemental reference).

Formula	mp-id / COD	Space group	E_{hull} (eV/at)	ΔICOHP/at. (eV)	Label
Ba_5P_4*	mp-17566	Pnma	0.0000	-0.9590	exobondic
$BaCl_2$*	mp-23199	Pnma	0.0068	-0.2244	exobondic
BaN_2*	mp-1102371	Cc	2.0568	-8.6248	exobondic
BaS_3*	mp-556296	P2_12_12	0.0436	-0.6233	exobondic
CaN*	mp-1058549	Fm-3m	0.3982	-4.9183	exobondic
Cs_2S*	mp-1077814	P6_3/mmc	0.0667	0.9554	endobondic
CsCl*	mp-573697	Fm-3m	0.0071	0.8830	endobondic
CsF*	mp-8455	Pm-3m	0.1088	3.8464	endobondic
K_2S*	mp-1022	Fm-3m	0.0000	-0.3359	exobondic
KCl*	mp-23289	Pm-3m	0.0776	0.7616	endobondic
KN_3	mp-636056	I4/mcm	1.0370	-5.9991	exobondic
Li_3N*	mp-2341	P6_3/mmc	0.0039	0.6165	endobondic
Mn_2O_7*	mp-28338	P2_1/c	0.3346	-1.4687	exobondic
MnO_2*	mp-510408	P4_2/mnm	0.0464	-0.5274	exobondic
Na_2Cl	mp-1077015	Cmmm	0.2391	-0.3965	exobondic
PbN_6*	mp-667338	Pnma	0.1093	-2.5257	exobondic
RbCl*	mp-23299	Pm-3m	0.0298	1.7924	endobondic
S_2N*	COD:4031496	P4_2nm	–	-0.9767	exobondic
SN*	COD:7017102	P2_1/c	–	-1.3963	exobondic
SrN*	mp-1058586	Fm-3m	0.4343	-4.5395	exobondic
SrS*	mp-10627	Pm-3m	0.2996	-0.1088	exobondic

H Endobondic Reactions

The subset of case-1 decomposition reactions ($A_aB_b \to a\,A + b\,B$, standard-state elemental references, balanced by `reaction_icohp.py`) labeled **endobondic** (ΔICOHP ≥ 0, products − reactants convention — note this is the sign *opposite* `reaction_icohp.py`'s own `delta_icohp_per_atom` column reported elsewhere in this project; see `delta.py`'s docstring).

Table S7: Decomposition-into-elements (case 1) reactions labeled **endobondic** (ΔICOHP $\geq$ 0, products $-$ reactants) among the 163 compounds of the `extension` campaign. Black * = stable experimental starting compound; red * = metastable experimental.

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	ΔICOHP/at. (eV)
0.3333 Be_3N_2 → Be + 0.3333 N_2	mp-1017550	0.2081	0.3960
$BeCl_2$ extcolorred* → Be + Cl_2	mp-23267	0.0087	1.8204
BeF_2 extcolorred* → Be + F_2	mp-15951	0.0004	3.1514
BeO extcolorred* → Be + 0.5 O_2	mp-7599	0.0153	2.7864
BeO → Be + 0.5 O_2	mp-1778	0.0070	4.4391
0.5 Cs_2S extcolorred* → Cs + 0.5 S	mp-1077814	0.0667	0.9554
CsCl extcolorred* → Cs + 0.5 Cl_2	mp-573697	0.0071	0.8830
CsF extcolorred* → Cs + 0.5 F_2	mp-8455	0.1088	3.8464
CsO_2 extcolorred* → Cs + O_2	mp-1441	0.0349	0.0666
CsO_2 → Cs + O_2	mp-1096936	0.0940	0.7743
KCl extcolorred* → K + 0.5 Cl_2	mp-23289	0.0776	0.7616
0.5 Li_2O* → Li + 0.25 O_2	mp-1960	0.0000	2.3366
0.5 Li_2O → Li + 0.25 O_2	mp-755894	0.0844	1.9481
0.5 Li_2S extcolorred* → Li + 0.5 S	mp-1125	0.0615	1.8505
0.5 Li_2S → Li + 0.5 S	mp-557142	0.1750	1.5052
0.3333 Li_3N extcolorred* → Li + 0.1667 N_2	mp-2341	0.0039	0.6165
LiCl extcolorred* → Li + 0.5 Cl_2	mp-22905	0.0243	2.8030
LiCl → Li + 0.5 Cl_2	mp-1185319	0.0039	2.5005
LiF* → Li + 0.5 F_2	mp-1138	0.0000	4.4673
LiF → Li + 0.5 F_2	mp-1009009	0.2875	4.5131
LiP_5 extcolorred* → Li + 5 P	mp-2412	0.0077	0.9043
LiP_5 → Li + 5 P	mp-32760	0.0963	0.8692
$MgCl_2$* → Mg + Cl_2	mp-23210	0.0000	0.1568
MgF_2 extcolorred* → Mg + F_2	mp-1072956	0.0025	0.0015
MgF_2 → Mg + F_2	mp-1062842	0.2636	0.0229
0.5 Rb_2O extcolorred* → Rb + 0.25 O_2	mp-1394	0.0032	0.7235
RbCl extcolorred* → Rb + 0.5 Cl_2	mp-23299	0.0298	1.7924
RbF extcolorred* → Rb + 0.5 F_2	mp-2064	0.0516	3.9244
RbF → Rb + 0.5 F_2	mp-11718	0.0000	1.9497

I Exobondic Reactions

The complementary **exobondic** subset (ΔICOHP $<$ 0), same convention and provenance as Appendix H.

Table S8: Decomposition-into-elements (case 1) reactions labeled **exobondic** (ΔICOHP $<$ 0, products $-$ reactants) among the 163 compounds of the `extension` campaign. Black * = stable experimental starting compound; red * = metastable experimental.

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	ΔICOHP/at. (eV)
0.2 Ba_5P_4 extcolorred* → Ba + 0.8 P	mp-17566	0.0000	-0.9590
$BaCl_2$ extcolorred* → Ba + Cl_2	mp-23199	0.0068	-0.2244
BaF_3* → Ba + 1.5 F_2	mp-1080821	0.0000	-0.8680
BaF_3 → Ba + 1.5 F_2	mp-1183303	0.0612	-0.2493
BaN_2 extcolorred* → Ba + N_2	mp-1102371	2.0568	-8.6248
BaO extcolorred* → Ba + 0.5 O_2	mp-1008500	0.0189	-0.3221
BaO → Ba + 0.5 O_2	mp-1950989	0.0434	-1.6129
BaS_3 extcolorred* → Ba + 3 S	mp-556296	0.0436	-0.6233

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	ΔICOHP/at. (eV)
0.3333 Be_3N_2* → Be + 0.3333 N_2	mp-18337	0.0000	-0.0746
0.3333 Be_3P_2* → Be + 0.6667 P	mp-567841	0.0000	-0.2111
0.3333 Be_3P_2 → Be + 0.6667 P	mp-1084771	0.1567	-0.1795
$BeCl_2$ → Be + Cl_2	mp-1172909	3.1688	-5.7807
BeF_2 → Be + F_2	mp-975644	2.5181	-0.6816
0.3333 Ca_3N_2 extcolorred* → Ca + 0.3333 N_2	mp-1047	0.0167	-4.4881
0.3333 Ca_3N_2* → Ca + 0.3333 N_2	mp-844	0.0000	-4.4374
0.3333 Ca_3N_2 → Ca + 0.3333 N_2	mp-1013524	0.5112	-4.4898
$CaCl_2$ extcolorred* → Ca + Cl_2	mp-23214	0.0011	-0.3199
$CaCl_2$ → Ca + Cl_2	mp-1120739	0.0422	-0.2208
CaF_2 extcolorred* → Ca + F_2	mp-10464	0.0346	-0.2227
CaF_2 → Ca + F_2	mp-554355	0.2403	-0.1814
CaN extcolorred* → Ca + 0.5 N_2	mp-1058549	0.3982	-4.9183
CaO extcolorred* → Ca + 0.5 O_2	mp-1079707	0.2098	-2.2127
CaO* → Ca + 0.5 O_2	mp-2605	0.0000	-0.9525
CaO → Ca + 0.5 O_2	mp-1181903	2.7281	-2.9971
CsN_3* → Cs + 1.5 N_2	mp-510557	0.0000	-2.2903
CsN_3 → Cs + 1.5 N_2	mp-1225892	0.0113	-1.6664
CsP_7* → Cs + 7 P	mp-1201790	0.0000	-0.2978
CsP_7 → Cs + 7 P	mp-1213206	0.0001	-0.2667
0.5 K_2O* → K + 0.25 O_2	mp-971	0.0000	-0.4357
0.5 K_2O → K + 0.25 O_2	mp-1223784	0.2000	-1.5638
0.5 K_2S* → K + 0.5 S	mp-1022	0.0000	-0.3359
KF_3 extcolorred* → K + 1.5 F_2	mp-1544557	0.0125	-1.1511
KF_3 → K + 1.5 F_2	mp-2749823	0.0119	-1.1505
KN_3 → K + 1.5 N_2	mp-636056	1.0370	-5.9991
KP extcolorred* → K + P	mp-1058315	1.0080	-2.1600
KP → K + P	mp-1057787	1.0092	-2.1366
$MgCl_2$ → Mg + Cl_2	mp-569626	0.1991	-0.0862
MgO extcolorred* → Mg + 0.5 O_2	mp-549706	0.1180	-1.4179
MgO → Mg + 0.5 O_2	mp-1009127	0.7484	-1.0763
MgS* → Mg + S	mp-1315	0.0000	-0.3401
MgS → Mg + S	mp-13032	0.0343	-0.4163
0.5 Mn_2O_7 extcolorred* → Mn + 1.75 O_2	mp-28338	0.3346	-1.4687
MnO_2 extcolorred* → Mn + O_2	mp-510408	0.0464	-0.5274
0.5 Na_2Cl → Na + 0.25 Cl_2	mp-1077015	0.2391	-0.3965
0.5 Na_2O* → Na + 0.25 O_2	mp-2352	0.0000	-1.3057
0.5 Na_2O → Na + 0.25 O_2	mp-1975297	0.0981	-1.5619
NaN_3 extcolorred* → Na + 1.5 N_2	mp-22003	0.0017	-1.5201
NaN_3 → Na + 1.5 N_2	mp-1065265	1.0162	-5.4248
NaS extcolorred* → Na + S	mp-409	0.0023	-0.4911
NaS → Na + S	mp-1180106	1.0061	-2.3999
PbN_6 extcolorred* → Pb + 3 N_2	mp-667338	0.1093	-2.5257
0.5 Rb_2O → Rb + 0.25 O_2	mp-2023482	0.0479	-0.7592
RbN_3 extcolorred* → Rb + 1.5 N_2	mp-581833	0.0193	-1.3478
RbN_3 → Rb + 1.5 N_2	mp-1219567	0.7966	-3.5636
RbP extcolorred* → Rb + P	mp-1179688	0.9879	-1.9315
RbP → Rb + P	mp-1058499	0.9882	-1.9104
RbS extcolorred* → Rb + S	mp-558071	0.0019	-0.3217
RbS → Rb + S	mp-1179684	0.9975	-2.7347
0.5 S_2N extcolorred* → S + 0.25 N_2	COD:4031496	–	-0.9767
SN extcolorred* → S + 0.5 N_2	COD:7017102	–	-1.3963
0.3333 Sr_3P_4* → Sr + 1.333 P	mp-14288	0.0000	-1.0472
0.3333 Sr_3P_4 → Sr + 1.333 P	mp-682953	0.0081	-1.0793
SrF_2 extcolorred* → Sr + F_2	mp-1102240	0.0646	-0.5421
SrF_2 → Sr + F_2	mp-1102911	0.0267	-0.1481

Reaction (balanced)	mp-id / COD	E_{hull} (eV/at)	ΔICOHP/at. (eV)
SrN extcolorred* → Sr + 0.5 N_2	mp-1058586	0.4343	-4.5395
SrO_2 extcolorred* → Sr + O_2	mp-726705	0.0096	-1.9205
SrO_2 → Sr + O_2	mp-2763179	0.0110	-1.7522
SrS extcolorred* → Sr + S	mp-10627	0.2996	-0.1088
TiO_2 extcolorred* → Ti + O_2	mp-1439	0.0045	-0.6461
TiO_2 extcolorred* → Ti + O_2	mp-9173	0.0575	-0.6509
TiO_2 extcolorred* → Ti + O_2	mp-430	0.0395	-0.6267